Wealth Inequality in Korea: Limited Participation and Idiosyncratic Returns*

Ilhwan Song†

August 2026

Abstract

Wealth in Korea is highly concentrated: the top 10% of households hold 44% of total wealth, with a Gini coefficient of 0.61. This paper develops a heterogeneous-agent general equilibrium model that combines limited asset market participation with persistent idiosyncratic returns to wealth to account for this skewed distribution. Calibrated to Korean microdata, the model replicates the observed wealth and income distributions. Counterfactual policy experiments uncover a key trade-off: expanding access to risky asset markets reduces the wealth Gini by 2 percentage points without distorting the macroeconomy, while traditional labor and wealth taxes are ineffective at reducing wealth inequality through general equilibrium channels but generate aggregate welfare gains through redistribution.

Keywords: wealth inequality, limited market participation, idiosyncratic returns
JEL Classification: D14, D31, E21, G11

---

*The author acknowledges support from the 2025 Research Fund of the University of Seoul, South Korea. I thank the editor, the associate editor, and two anonymous referees for comments that substantially improved the paper.

†Business School, University of Seoul, ilhwansong@uos.ac.kr

1 Introduction

The concentration of wealth at the top has become a defining feature of many economies. In the United States, the richest 10 percent of households hold over 75 percent of total wealth, with the top 1 percent alone controlling more than 35 percent (Xavier, 2021). Similar patterns are found in Korea, where widening gaps between homeowners and non-homeowners have further amplified inequality (Park et al., 2024). Understanding the mechanisms behind these skewed wealth distributions remains a central challenge in macroeconomics.

Classic quantitative models of inequality, such as Aiyagari (1994), Huggett (1993), and Krusell and Smith (1998), emphasize uninsurable idiosyncratic income risk, but they fail to reproduce the extreme concentration observed in the data. Recent work, such as Benhabib et al. (2019), Hubmer et al. (2021), Fagereng et al. (2020) highlights the importance of return heterogeneity—persistent differences in returns across households. These studies show that return heterogeneity plays a more significant role than heterogeneity in patience, earnings, or tax progressivity. Yet, they typically impose reduced-form return processes or abstract from participation frictions in asset markets.

	Top 1%	Top 5%	Top 10%	Gini Coefficient
US Data [a]	37	65	76	0.85
Krusell and Smith (1998)	3	11	19	0.25
Korean Data [b]	—	—	44	0.61

Table 1: Distribution of Wealth: Data and Benchmark Models

NOTES: [a] Xavier (2021) [b] 2024 Survey of Household Finances and Living Conditions; top 1% and 5% shares not reported in published statistics.

Another strand of research emphasizes limited participation in financial markets. Guvenen (2009) shows that differences between stockholders and non-stockholders help explain inequality, but his framework abstracts from idiosyncratic returns. Similarly, studies of the housing market highlight that homeownership is a key margin shaping cross-country differences in wealth inequality, as housing remains the dominant risky asset for many households. According to Causa et al. (2019), there is a strong negative relationship between homeownership rates and wealth inequality. They emphasize that participation in the housing market plays a crucial role in explaining cross-country differences in wealth inequality, using micro-level household data from OECD countries. In the context of the Korean economy, Park et al. (2024) argue that the recent increase in wealth inequality can largely be attributed to the widening gap between homeowners and non-homeowners. Therefore, participation in the housing market is a key channel through which housing price dynamics impact wealth inequality.

While the household's decision to participate in risky asset markets is in itself an important research question, this paper takes the participation margin as given. This allows us to focus specifically on the quantitative importance of idiosyncratic returns and the existing division between participants and non-participants in shaping wealth inequality.

This paper integrates these two strands by developing a parsimonious general equilibrium model that incorporates both (i) limited participation in risky asset markets and (ii) persistent idiosyncratic returns to wealth. Calibrating the model to Korean data, we show that these features are essential to replicate the observed concentration of wealth. We then use the model to evaluate redistributive policies, comparing the effects of labor income taxation, wealth taxation, and policies expanding risky asset participation.

This paper makes three contributions. First, we extend the Aiyagari-Bewley-Huggett tradition by combining limited participation in risky asset markets (Guvenen, 2009) with persistent idiosyncratic returns (Fagereng et al., 2020). This joint framework captures both access frictions and heterogeneity in returns. Second, we calibrate the model to Korean microdata, highlighting the central role of housing and risky asset participation margins in shaping wealth inequality. Third, we compare the effects of labor income taxes, taxes on returns to wealth, and promoting risky market participation policies. The results show that increasing access to risky asset markets can alleviate wealth inequality. In contrast, while labor and wealth taxes are more effective in enhancing overall welfare, they have a limited effect on wealth inequality.

Scope and limitations. This paper proxies risky asset market participation by the homeownership rate. In the model, the risky asset itself is a broad pool of household claims on productive capital—encompassing housing, real estate, equities, and business ownership—and homeownership serves as the empirical measure of who participates in this pool. The choice reflects a data-driven mapping suited to the Korean context: according to the 2024 SFLC, housing assets account for approximately 70% of household wealth in Korea, while financial assets account for only 18% and direct stock holdings only 9.8%, making homeownership a natural empirical proxy for risky-asset participation. This contrasts with prior work measuring participation through stock markets (Guvenen, 2009), where stocks are the dominant risky asset in the U.S. context. Modeling considerations also support treating the pooled household savings as productive capital: there is precedent for incorporating real estate into the production sector in the macroeconomic literature, ranging from the basic framework of Iacoviello (2005), in which real estate directly enters the production function, to Liu et al. (2013), who treat land as a distinct productive input alongside conventional capital. Our objective, however, is not to model housing per se but to use the homeownership margin as a tractable proxy for risky-asset participation. While tractable, this approach necessarily abstracts from housing-specific features such as illiquidity, mortgage leverage, and collateral constraints. The absence of a collateral channel affects both the baseline wealth distribution and the magnitude of the counterfactual results; Section 3 discusses these channels in more detail. In addition, participation is treated as exogenous, whereas in reality households endogenously choose whether to enter the housing market in response to financial frictions such as down-payment requirements and credit limits. Incorporating these features—endogenous participation and housing-specific frictions—would enrich the analysis but lies beyond the scope of this paper; we leave such extensions for future research. Despite these limitations, the model provides a tractable and internally consistent framework for quantifying the roles of return heterogeneity and participation margins in shaping Korea's wealth distribution, and for drawing first-order policy conclusions.

Related literature. This paper relates to previous studies about wealth distribution by using quantitative macroeconomic models with uninsurable idiosyncratic income risks. Aiyagari (1994), Bewley (1977), and Huggett (1993) provide the fundamental theory and workhorse models of wealth inequality. Krusell and Smith (1998) extend the workhorse models with aggregate productivity risks. However, the workhorse models of wealth inequality failed to match the highly skewed wealth distribution from the data.

In addition, many recent studies focus on return heterogeneity in order to explain the highly concentrated wealth in the top wealth groups. In empirical studies, Fagereng et al. (2020) find that return heterogeneity is mainly caused by differences in the allocation of wealth between safe and risky assets, the positive correlation of returns with wealth, and persistent individual returns to wealth from Norway's administrative data. Bach et al. (2020) argue that return heterogeneity heavily depends on the different portfolio selections of individuals from Swedish administrative data. Also, they state that return heterogeneity mostly explains the historical increase in top wealth shares.

In quantitative studies about return heterogeneity, Benhabib et al. (2019) develop a quantitative life-cycle model with return heterogeneity to explain both the highly skewed U.S. wealth distribution and the observed degree of social mobility. Since, however, it is a partial equilibrium model, it does not analyze the general equilibrium effects of taxation on wealth inequality. Also, it does not incorporate limited market participation or heterogeneity in asset allocation choices. Hubmer et al. (2021) build on a Bewley-Huggett-Aiyagari model with return heterogeneity and examine the roles of tax progressivity, asset returns, and wage inequality. However, the model does not feature limited participation in risky asset markets. By incorporating limited market participation and general equilibrium taxation effects, this paper can capture how wealth inequality is shaped not only by return heterogeneity or taxation policy, but also by risky asset participation margins.

Lastly, this paper connects to studies about individuals' limited participation in risky asset markets like Basak and Cuoco (1998) and Guvenen (2009). Guvenen (2009) studies the large wealth inequality among households by using a two-agent macroeconomic model with limited participation in stock markets. There are two types of individuals: non-stockholders and stockholders. The former only accesses risk-free bond markets and the latter accesses both risk-free bond and stock markets. Guvenen (2009) finds that the high elasticity of inter-temporal substitution is the key factor to generate return heterogeneity and hugely concentrated wealth distribution. However, since Guvenen (2009) focuses on the asset pricing puzzle, it generates only a two-group wealth split between stockholders and non-stockholders without continuous within-group heterogeneity, due to the absence of idiosyncratic uninsurable income risks and return shocks to wealth.

We follow the same conceptual framework of dividing households by their access to risky returns, but adopt a different empirical measure of participation suited to the Korean context. Guvenen (2009) sets the U.S. stock market participation rate to 20%, deliberately capturing households that hold risky assets in meaningful quantities rather than nominal stockholders, noting that the top 20% of U.S. households have consistently held more than 98% of stocks. The appropriate participation margin thus depends on the asset through which households hold risky wealth in meaningful quantities. In Korea, that asset is housing rather than equity: housing constitutes 70% of household wealth, whereas stocks account for only 9.8% of financial assets (SFLC 2024), and housing is the central driver of Korean wealth inequality (Park et al., 2024). Although a non-trivial share of Korean individuals hold some listed equity, these holdings are small in scale relative to housing, so homeownership (60.7%) is the empirical margin that captures meaningful risky-asset participation. The model's risky asset itself is a broad pool of household claims on productive capital—encompassing housing, real estate, equities, and business ownership—rather than housing alone.

The remainder of the paper is organized as follows. Section 2 presents the model. Section 3 presents the calibration, quantitative results, and policy experiments. Section 4 concludes.

2 Model

Our model aims to generate a realistic wealth distribution by using the heterogeneity of returns to wealth and the limited participation in the risky asset market. The work most similar to ours is by Hubmer et al. (2021); however, our approach differs primarily by incorporating the concept of limited participation in the risky asset market. Our approach of incorporating limited participation follows a tradition in macroeconomics (Guvenen, 2009) that separates households into stockholders and non-stockholders to explain key macroeconomic phenomena.

By employing the frameworks, we can incorporate heterogeneous portfolio allocation between risk-free and risky assets into the workhorse macroeconomic model. One type of household allocates its wealth to risky assets, while another type invests its savings in risk-free assets. This approach allows for an evaluation of how the differing decisions of households to participate in the risky asset market influence the shape of wealth distribution. However, the model does have limitations. Portfolio selection is not determined endogenously; rather, it is assigned exogenously based on household type. We model the share of participants exogenously, set to 60.7% based on the homeownership rate. In reality, this rate is an equilibrium object shaped by financial frictions such as down-payment requirements and credit constraints; we hold it fixed not because it is immune to such frictions but in order to isolate the consequences of a given participation level for the wealth distribution. This abstraction is empirically supported by the strong regularity in Korea, where the homeownership rate has remained remarkably stable for the past eight years, making a fixed rate a reasonable and tractable approximation for the long-run analysis we undertake.

The model is set in a closed economy. Three return variables appear in the model: the risk-free rate r_f, the common risky return $\underline{r}_t$, and the idiosyncratic return component $r_{i,t}$. The risk-free rate r_f is set exogenously to the long-run average real yield on Korean Treasury bonds. The common return $\underline{r}_t$ is endogenously determined each period to ensure that aggregate returns paid to risky-asset holders equal the rental revenue of mutual funds, Equation (15). The idiosyncratic component $r_{i,t}$ follows the exogenous AR(1) process in Equation (2).

Fagereng et al. (2020) find that persistent idiosyncratic return heterogeneity is a key factor in generating a highly skewed wealth distribution by examining Norway's administrative data. Since the workhorse macroeconomic models failed to match wealth shares of the top wealth groups in the wealth distribution, we include persistent idiosyncratic return components for risky asset holdings in the model in order to resolve the problem of the workhorse model.

In this model, there are four economic agents: households, representative firms, mutual funds, and the government.

Environment. There is a stochastic shock to idiosyncratic labor productivity, $e_{i,t}$, which follows a first-order Markov process with the transition probabilities, $\pi_{e_t,e_{t+1}}$. The idiosyncratic labor productivity satisfies the law of large numbers.

$$\log e_{i,t} = \rho_e \log e_{i,t-1} + \epsilon_{e,i,t}, \quad \epsilon_{e,i,t} \sim N(0, \sigma_e) \tag{1}$$

where ρ_e is the persistence of labor productivity shock, ϵ_e is a transitory shock, and σ_e is the standard deviation of shocks to the individual labor productivity.

There is an additional stochastic shock to an idiosyncratic return component, $r_{i,t}$, which follows a first-order Markov process given by the transition probabilities, $\pi_{r_t,r_{t+1}}$:

$$r_{i,t} = \rho_r r_{i,t-1} + \epsilon_{r,i,t}, \quad \epsilon_{r,i,t} \sim N(0, \sigma_r) \tag{2}$$

where ρ_r is the persistence of previous shock for returns, ϵ_r is a transitory shock for returns, and σ_r is the standard deviation of shocks to the individual returns to risky assets. We assume that the individual asset returns to risky assets are the sum of a common return component, $\underline{r}_t$, and an individual specific return component, $r_{i,t}$.

The idiosyncratic return process warrants a justification distinct from that of the labor productivity process. For labor productivity, we adopt the AR(1) specification that has been standard since Aiyagari (1994), with persistence and dispersion in the range commonly used in the literature. For idiosyncratic returns, by contrast, the literature has not converged on a single canonical functional form: Benhabib et al. (2019) and Xavier (2021) model persistence as a permanent return *type* that is fixed over an agent's horizon, Hubmer et al. (2021) generate persistence through *scale dependence* in which expected returns rise with accumulated wealth, while Fagereng et al. (2020), using Norwegian administrative data, empirically decompose individual returns into a large permanent component—accounting for roughly 60% of the explained variation—and a transitory part. Bach et al. (2020) document similarly substantial persistence using Swedish data. The common thread across these studies, which span different countries and data sources, is that returns to wealth exhibit strong and persistent heterogeneity.

We model idiosyncratic returns as a persistent AR(1) process, discretized via Tauchen (1986). This specification nests the permanent-type approach of Benhabib et al. (2019) and Xavier (2021) as the limiting case $\rho_r \to 1$, while additionally permitting transitory variation, and is therefore a transparent and comparatively general choice. We defer discussion of how the persistence and dispersion parameters are disciplined to the calibration section, where the identification of these parameters is addressed in detail.

For convenience in notation, let us define $s_{i,t}$ as the pair of idiosyncratic shocks $(e_{i,t}, r_{i,t})$. Let $\pi_{s_t,s_{t+1}}$ denote the transition matrix of a pair of idiosyncratic shocks.

Household. There is a continuum of infinitely lived individuals who fall into two categories: participants in the risky asset market and non-participants. The population share of participants is represented by λ, while the share of non-participants is denoted as $1-\lambda$. Each individual, i, in both types maximizes her lifetime utility using a subjective discount factor, β, through her consumption choices, $c_{i,t}$, and labor, $l_{i,t}$.

$$\max \sum_{t=0}^{\infty} \mathbb{E}_0 \beta^t [u(c_{i,t}, l_{i,t})] \tag{3}$$

We use the period utility derived from the GHH preference framework, as described by Greenwood et al. (1988), which exhibits no income effect on hours worked.[1]

$$u(c_{i,t}, l_{i,t}) = \frac{1}{1-\gamma} \left(c_{i,t} - \psi \frac{l_{i,t}^{1+\theta}}{1+\theta} \right)^{1-\gamma} \tag{4}$$

where γ is constant relative risk aversion, θ is inverse of Frisch elasticity, ψ is labor disutility weight.

The participants in the risky asset market face idiosyncratic risks associated with the

[1] While our model incorporates endogenous labor supply to accommodate the labor income tax experiment, the primary focus of this paper is on wealth inequality and the mechanisms through which return heterogeneity and participation margins shape the wealth distribution. This focus motivates our choice of GHH preferences on both empirical and analytical grounds.

Empirically, Ferraro and Valaitis (2025) document, using PSID data for 2001–2015, that market hours worked are approximately flat across the wealth distribution in the United States—wealthy households work nearly as much as less affluent ones. They explicitly note that this finding is "a standing challenge for standard heterogeneous-agent macro models," in which wealthier households work fewer hours due to the income effect on labor supply. This evidence suggests that CRRA separable preferences, which generate a strong negative wealth–labor supply relationship through the income effect, would misrepresent the empirical labor supply behavior of wealthy households and, more importantly for the purposes of the present paper, distort the replication of wealth distribution moments.

Analytically, the wealth inequality literature typically abstracts from labor supply entirely (Aiyagari, 1994; Huggett, 1993; Krusell and Smith, 1998; Hubmer et al., 2021), as asset income tends to dominate labor income for wealthy households and endogenous labor supply with income effects can contaminate the wealth accumulation mechanism. Since our model requires endogenous labor supply for the tax experiments, GHH preferences preserve the spirit of this literature by ensuring that labor supply responds only to wages and productivity rather than to asset levels. We acknowledge that CRRA separable preferences could affect the quantitative magnitude of the policy results, and leave this robustness analysis for future work.

returns on their asset holdings. As a result, they need to make decisions about how much to invest in risky assets based on their expectations of future wealth returns. Since each participant is small in scale, they take wages and common return components as given. Each participant must decide how to allocate their resources between risky asset savings, $a_{i,t+1}$, and consumption, $c_{i,t}$, as well as how many hours to dedicate to labor, $l_{i,t}$. They have to pay labor income taxes on their earnings and wealth taxes on asset returns. Each participant is subject to the following budget constraint.

$$c_{i,t} + a_{i,t+1} = [1 + (1 - \tau_w)(\underline{r}_t + r_{i,t})]a_{i,t} + (1 - \tau_l)w_t e_{i,t} l_{i,t} + T_t \tag{5}$$

where $a_{i,t}$ is risky asset holdings, w_t represents wage per efficient unit, $\underline{r}_t$ is common return component to risky asset holdings, $r_{i,t}$ is idiosyncratic part of returns to risky asset holdings, τ_w is wealth tax rate on returns to wealth, τ_l is labor income tax rate, $e_{i,t}$ is individual labor productivity, T_t is a lump-sum transfer from the government.

In this model, the 'wealth tax' is modeled as a tax on returns to wealth rather than a levy on asset holdings, thereby directly reducing the effective return on capital income. Similarly, the 'labor tax' in the model should be understood as a labor income tax imposed on earnings from employment.

In recursive form, let V^p represent the value for the participants.

$$\begin{aligned} V^p(s,a) = \max_{c,a',l} \; & u(c,l) + \beta \mathbb{E}_{s'|s}[V^p(s',a')] \\ \textbf{s.t. } & c + a' = [1 + (1-\tau_w)(\underline{r} + r)]a + (1-\tau_l)wel + T \\ & c \geq 0, \; a' \geq 0, \; l \in [0,1] \end{aligned} \tag{6}$$

$\mathcal{C}^p(s,a)$ is associated policy function for consumption, $\mathcal{A}(s,a)$ is associated policy function for saving a', and $\mathcal{L}^p(s,a)$ is associated policy function for labor supply.

Let μ_p be the distribution of participants.

$$\mu_p'(s',a') = \sum_{s \in S} \Pi(s',s) \; \mathbb{I}_{[\mathcal{A}(s,a)=a']} \; \mu_p(s,a) \tag{7}$$

where μ_p' is the next period distribution, $\Pi(s',s)$ is transition matrix over idiosyncratic labor productivity and return shocks, $\mathbb{I}$ is an index function where $\mathbb{I}$ equals one if and only if the condition, $\mathcal{A}(s,a) = a'$, holds.

The non-participants in the risky asset market can only save in a risk-free asset and do not face any uncertainty regarding returns on their wealth. Since each non-participant is small in scale, they take wage rates and risk-free rates as given. Every period, each non-participant decides how to allocate their resources between risk-free savings, denoted as $b_{i,t+1}$, and consumption, $c_{i,t}$, given cash on hand. They also need to determine how many hours to dedicate to labor $l_{i,t}$. Like participants, they need to pay labor taxes on their labor incomes and wealth taxes on risk-free asset returns. Each non-participant is subject to the following budget constraint.

$$c_{i,t} + b_{i,t+1} = [1 + (1 - \tau_w) r_f] b_{i,t} + (1 - \tau_l) w_t e_{i,t} l_{i,t} + T_t \tag{8}$$

where $b_{i,t}$ is risk-free asset holdings, w_t represents wage per efficient unit, r_f is risk-free rate, τ_w is wealth tax rate on returns to asset holdings, τ_l is labor income tax rate, $e_{i,t}$ is individual labor productivity, T_t is a lump-sum transfer.

In recursive form, let V^n symbolize the value for the non-participants.

$$\begin{aligned} V^n(s, b) = \max_{c, b', l} \; & u(c, l) + \beta \mathbb{E}_{s'|s}[V^n(s', b')] \\ \textbf{s.t. } & c + b' = [1 + (1 - \tau_w) r_f] b + (1 - \tau_l) wel + T \\ & c \geq 0, \; b' \geq 0, \; l \in [0, 1] \end{aligned} \tag{9}$$

$\mathcal{C}^n(s, b)$ is associated policy function for consumption, $\mathcal{B}(s, b)$ is associated policy function for saving b', and $\mathcal{L}^n(s, b)$ is associated policy function for labor supply.

Let μ_n be the distribution of non-participants.

$$\mu_n'(s', b') = \sum_{s \in S} \Pi(s', s) \; \mathbb{I}_{[\mathcal{B}(s,b) = b']} \; \mu_n(s, b) \tag{10}$$

where μ_n' is the next period distribution, $\mathbb{I}$ is an index function where $\mathbb{I}$ equals one if and only if the condition, $\mathcal{B}(s, b) = b'$, holds.

Representative firms. Representative firms borrow capital from mutual funds and hire labor from both types of households for production. These firms pay a rental rate for the capital, compensate for the amount of depreciated capital, and cover the wage bill for hired labor. The production function adopted by the firms is a Cobb-Douglas function, which incorporates capital stock, K_t, and labor hired, L_t. The firms maximize their profit, Π_t, while deciding on the amounts of capital and labor to hire. Since each representative firm is relatively small in scale, they take the rental rates r_t^* and wages w_t as given. The profit maximization problem can be expressed as follows:

$$\Pi_t = \max_{K_t, L_t} \; K_t^{\alpha} L_t^{1-\alpha} - (r_t^* + \delta) K_t - w_t L_t \tag{11}$$

where α represents income share of capital stock, δ is the depreciation rate.

The optimal conditions for the profit maximization problem satisfy two equations:

$$r_t^* = \alpha K_t^{\alpha-1} L_t^{1-\alpha} - \delta \tag{12}$$

$$w_t = (1-\alpha) K_t^{\alpha} L_t^{-\alpha} \tag{13}$$

These equations help determine the optimal levels of capital and labor that firms should employ to maximize their profits.

Mutual funds. Risk-neutral mutual funds issue two types of financial instruments: risky assets, a_t, and risk-free assets, b_t. These mutual funds collect contributions from households through these instruments. They pool all savings into new capital stock, K_{t+1}, for the next period and then rent this capital stock to representative firms. The firms pay rents, r_t^*, to the mutual funds for borrowing capital. In turn, the mutual funds distribute all earnings from renting out capital stock to both types of households. Households that invest in risk-free assets receive fixed and risk-free interest rates, r_f, while those who invest in risky assets receive both common returns, $\underline{r}_t$, and an idiosyncratic return component, $r_{i,t}$.

$$K_{t+1} = \lambda \int a_{i,t+1} d\mu_p + (1-\lambda) \int b_{i,t+1} d\mu_n \tag{14}$$

$$r_t^* K_t = \lambda \int (\underline{r}_t + r_{i,t}) a_{i,t} d\mu_p + (1-\lambda) \int r_f b_{i,t} d\mu_n \tag{15}$$

In this framework, the common return component of the risky asset, $\underline{r}_t$, is not exogenously given but is endogenously determined to satisfy Equation (15), which ensures that the mutual fund's aggregate return is consistently allocated across all participants. This highlights the model's core mechanism of heterogeneous claims on a single capital pool.

Government. The government collects labor and wealth taxes from both types of households and transfers the tax revenue to them in a lump-sum fashion. The government ensures that it satisfies the balanced budget condition each period.

$$\lambda \int T_t d\mu_p + (1-\lambda) \int T_t d\mu_n \tag{16}$$
$$= \lambda \int [\tau_w(\underline{r}_t + r_{i,t})a_{i,t} + \tau_l w_t e_{i,t} l_{i,t}] d\mu_p + (1-\lambda) \int [\tau_w r_f b_{i,t} + \tau_l w_t e_{i,t} l_{i,t}] d\mu_n$$

Market clearing. There are two markets to clear: the capital market and labor market. To clear all markets, we need conditions for capital and labor market clearance.

$$K_t = \lambda \int \, a_{i,t} \, d\mu_p + (1-\lambda) \int \, b_{i,t} \, d\mu_n \tag{17}$$

$$L_t = \lambda \int \, e_{i,t} l_{i,t} \, d\mu_p + (1-\lambda) \int \, e_{i,t} l_{i,t} \, d\mu_n \tag{18}$$

Recursive Equilibrium. A recursive competitive equilibrium is a set of functions, $V^p(s,a)$, $V^n(s,b)$, $\mathcal{C}^p(s,a)$, $\mathcal{C}^n(s,b)$, $\mathcal{L}^p(s,a)$, $\mathcal{L}^n(s,b)$, $\mathcal{A}(s,a)$, $\mathcal{B}(s,b)$, distributions, $\mu_p(s,a)$, $\mu_n(s,b)$ prices, r^*, $\underline{r}$, w, aggregate capital K, aggregate labor L.

1. Given the prices and the value functions, the household decision rules, $\mathcal{C}^p(s,a)$, $\mathcal{C}^n(s,b)$, $\mathcal{L}^p(s,a)$, $\mathcal{L}^n(s,b)$, $\mathcal{A}(s,a)$, $\mathcal{B}(s,b)$, satisfy each agent's problems, Equations (6) and (9).

2. Given the prices and the decision rules, the household's value functions $V^p(s,a)$, $V^n(s,b)$, satisfy each agent's problems, Equations (6) and (9).

3. $\mu_p(s,a)$, $\mu_n(s,b)$ are fixed points of Equations (7) and (10).

4. Factor prices are competitively determined by Equations (12) and (13).

5. Mutual funds rules, Equations (14) and (15), and government balanced budget, Equation (16), are satisfied.

6. All markets clear by Equations (17) and (18).

3 Quantitative Analysis

3.1 Calibration

We calibrate the model for Korea by collecting wealth and income distribution data from the 2024 Survey of Household Finances and Living Conditions (SFLC), which was conducted by Statistics Korea, the Financial Supervisory Service, and the Bank of Korea. We categorize the parameters into two groups. The first group consists of parameters that can be set directly, based on estimates derived from the data and values that are commonly found in the literature. Table 2 presents these values along with their interpretations. The second group includes parameters that are specific to our model. These parameters are jointly calibrated to match the wealth and income inequality moments observed in the data. Table 3 and Table 4 list these parameters and the corresponding model fit.

	Value	Description
Preference		
γ	2.0	Constant relative risk aversion
β	0.96	Discount factor
θ	1.0	Inverse of Frisch elasticity
ψ	6.0	Labor disutility weight
Production		
α	0.36	Capital income share
δ	0.08	Capital depreciation
Tax rate		
τ_l	0.066	Labor income tax rate
τ_w	0.00	Wealth tax rate
Financial market		
λ	0.607	Share of participants
r_f	0.01	Risk-free rate

Table 2: Parameters Set Externally

The constant relative risk aversion coefficient, denoted as γ, is set at 2.0, while the inverse of Frisch elasticity, θ, is 1.0. The discount factor, β, is 0.96, considering that the model period is annual. These values are commonly used in the literature. The labor disutility weight, ψ, is set at 6.0 to align with the share of working hours relative to available hours for wage workers, which is 0.36 according to the Survey on Labor Conditions by Employment Type (Korean Ministry of Employment and Labor). For the factor shares parameter, α, and the depreciation rate, δ, we use typical values found in the literature: 0.36 and 8%, respectively.

The labor income tax rate, τ_l, is set at 6.6%, following the effective labor income tax rate for Korean wage earners reported by the National Assembly Budget Office (2023 Korean Taxation).[2] The wealth tax rate, τ_w, is set to 0.0% as a theoretical benchmark. Korea's

[2]Korea's effective labor income tax rate is among the lowest in the OECD, reflecting the country's extensive employment income deductions and tax credits (OECD Taxing Wages, 2025). Note that combining

wealth-related taxes are primarily levied on asset values (property tax and comprehensive real estate tax) rather than on returns to wealth, making a direct mapping to τ_w in the model conceptually ambiguous; we therefore treat the 10% wealth tax experiment as a hypothetical policy scenario designed to illustrate the general equilibrium effects of taxing asset returns.

The risk-free rate, r_f, is set to 1.0%, consistent with the long-run (2000–2025) average real yield on Korean Treasury bonds, which ranges from roughly 0.8% to 1.5% across maturities of one to ten years (nominal yield to maturity minus CPI inflation). The share of participants in risky asset markets is 60.7%, based on the homeownership rate of the 2024 Korea Housing Survey. In Korea, housing is the dominant component of household risky assets, with stock holdings playing a minor role. According to the SFLC 2024, the share of housing assets is 70%, while the share of financial assets is 18%. Among financial assets, deposits are the most preferred investment method at 87.3%, whereas households show a preference for stocks at only 9.8%. Accordingly, the homeownership rate serves as the natural empirical proxy for participation in risky-asset markets in the Korean context—in contrast to the U.S. setting, where stock market participation (Guvenen, 2009) is the more natural margin.

The model's risky asset is a broad pool of household claims on productive capital, encompassing housing, real estate, equities, and business ownership, and homeownership serves as the empirical measure of who participates in this pool rather than an attempt to model housing as a distinct asset class. Treating this pool as productive capital follows a common modeling choice in macroeconomics, where housing or real estate is incorporated into the production sector—most directly in the basic model of Iacoviello (2005), where real estate enters the production function, and in models such as Liu et al. (2013), which include land as a distinct productive input. The approach abstracts from housing-specific characteristics such as illiquidity, mortgage leverage, and collateral constraints, which shape both the baseline wealth distribution and the response to participation-expanding policies; we return to these channels in Section 3.

Four parameters have been jointly selected to target moments related to wealth and income distribution. These moments include the wealth Gini coefficient, the income Gini coefficient, and the wealth shares for the top 10%, 20%, 30%, 40%, and 50% of the population, as well as the income shares for the same groups. The persistence of labor productivity shocks, ρ_e, is set at 0.90, and the persistence of return shocks, ρ_r, is set at 0.97. The

personal income tax with employee social security contributions raises the total burden to approximately 16.3% for a single average-wage earner, but since social insurance contributions are associated with future benefit entitlements and are not redistributed as lump-sum transfers in the model, we use the personal income tax rate alone as the basis for τ_l.

	Value	Description
ρ_e	0.90	Persistence, labor productivity
ρ_r	0.97	Persistence, returns to wealth
σ_e	0.13	Shock innovations, labor productivity
σ_r	0.002	Shock innovations, returns to wealth

Table 3: Parameters Set Internally

	Top 10%	Top 20%	Top 30%	Top 40%	Top 50%	Gini Index
Wealth Share (%)						
Data	44	63	75	84	90	0.61
Model	42	62	75	85	91	0.61
Income Share (%)						
Data	24	39	52	63	72	0.32
Model	24	40	52	63	72	0.32

Table 4: Wealth and Income Distribution

volatility of shock innovations for labor productivity, σ_e, and returns to risky assets, σ_r, are 0.13 and 0.002, respectively. Overall, the model performs well in the moment-matching exercise. For wealth inequality metrics, the model shows a top 10% wealth share of 42% and a wealth Gini index of 0.61, both of which are close to the actual data, 44% and 0.61. In terms of income inequality, the model presents a top 10% income share of 24% and an income Gini index of 0.32, closely aligned with the data. Although top 1% and top 5% shares are not targeted in the calibration—the published SFLC statistics report wealth shares only at the decile level—the model also matches independent estimates of these upper-tail moments well; the comparison is reported in Appendix B.

The persistence and dispersion of the idiosyncratic return process, ρ_r and σ_r, deserve particular comment. Identifying these parameters requires linked, household-level panel data on returns to wealth—the administrative tax-registry records exploited by Fagereng et al. (2020) and Bach et al. (2020) for Norway and Sweden. Aggregate, regional, or sectoral return series cannot recover these objects: they average over households and so identify between-group rather than within-household dispersion and persistence. Such individual-level records are not accessible to researchers in Korea, and the survey panels that are available, such as the SFLC, measure asset stocks rather than the value-and-flow decomposition needed to construct individual returns. Facing the same constraint, Benhabib et al. (2019) discipline the return process by targeting wealth-distribution moments, and we follow that approach. The calibrated persistence ($\rho_r = 0.97$) is consistent with the large permanent component documented in the individual-level studies above, and Appendix E confirms that lower persistence moves the model away from the data—so the parameter is pinned down by the wealth distribution rather than chosen freely.

Moment	Model
Capital stock	3.44
Labor supply	0.36
Wage	1.88
Rental rate (%)	2.96
Common return for risky asset (%)	3.07
Lump-sum transfer	0.04

Table 5: Model Statistics

In the steady state, the baseline model economy accumulates a capital stock of 3.44. The aggregate labor supply is 0.36. The wage rate per efficient unit of labor is 1.88, and the capital rental rate is 2.96%. The common return component for risky assets is 3.07%, endogenously determined by the mutual fund's market-clearing condition (Equation 15).

Appendix C decomposes the model's concentration by channel. Two results are worth noting. First, return heterogeneity and limited participation each make a large and approximately independent contribution to wealth concentration—removing either lowers the top 10% wealth share by 6 percentage points—so neither is silently absorbing the role of the other. Second, both channels leave the income distribution essentially unchanged, confirming that the labor-productivity parameters govern income concentration while the return and participation margins specialize in wealth concentration.

3.2 Counterfactual Analysis of Policy Instruments

We examine the impact of various policy instruments on wealth inequality, using the distribution derived from the model. We consider three policy scenarios. First, we propose a 10 percentage point (pp) increase in the labor income tax on earnings from employment, raising the rate from 6.6% to 16.6%. Second, we suggest implementing a 10% wealth tax on returns from asset holdings. Lastly, we propose a policy aimed at increasing participation in the risky asset market by 10 percentage points, raising the participation rate from 60.7% to 70.7%. For each scenario, we find a new steady state and compare it to the baseline economy. This analysis reveals the general equilibrium effects of the new policy instruments on wealth and income inequality.

To ensure comparability across policy instruments, all three experiments adopt the same shock magnitude of 10 percentage points. While these represent large policy shifts, they are chosen to clearly illustrate the directional effects of each instrument and to facilitate direct comparison across policies. A shift of this magnitude is also not implausible over the longer horizon. Under current policy, Korea's rapid population aging is projected to generate substantial fiscal pressure: the government's Third Long-Term Fiscal Projection (Ministry of Economy and Finance, 2025) places the government-debt-to-GDP ratio at between 133% and 173% by 2065, depending on demographic and growth assumptions. Closing fiscal gaps of this scale would require sizable revenue adjustment, so a 10 percentage point increase in the labor income tax lies within the range of adjustment that demographic pressure may eventually necessitate. For a more robust analysis, Appendix D examines marginal adjustments in policy instruments, specifically a 1 percentage point increase in tax rates and a 1 percentage point rise in participation rates. As confirmed in the robustness checks (Appendix D), these qualitative findings are not sensitive to the size of the policy shock: even under marginal adjustments, the main implications remain virtually unchanged.

3.2.1 Labor Income Tax

In this scenario, the government raises the labor income tax by 10 percentage points (from 6.6% to 16.6%) and distributes the additional tax revenues equally to individuals in a lump-sum manner. This tax increase introduces a larger labor tax wedge into the households' labor-leisure condition, discouraging individuals from supplying labor.

$$\frac{u_2(c,l)}{u_1(c,l)} = -(1-\tau_l)we \tag{19}$$

As a result of this taxation, aggregate labor decreases by 0.05, from 0.36 to 0.31. Additionally, households save less because their after-tax labor earnings decline, especially impacting those with high labor incomes but low wealth. This is a crucial consideration when formulating new policies, particularly for the younger generation, who have higher productivity levels but lack opportunities to accumulate wealth. Labor income taxes therefore hinder this high-productivity younger generation from building wealth effectively. Consequently, the overall capital stock falls from 3.44 to 2.87, which leads to an increase in the rental rate from 2.96% to 3.30%. These higher rental rates raise the common return on risky assets from 3.07% to 3.31%, exacerbating wealth inequality compared to the baseline economy. The wealth Gini index rises from 0.61 to 0.67, and the top 10% wealth share increases from 42% to 47%.

However, the labor income tax alleviates income inequality by directly decreasing the

Moment	Baseline	Labor tax
Capital stock	3.44	2.87
Labor supply	0.36	0.31
Wage	1.88	1.85
Rental rate (%)	2.96	3.30
Common return for risky asset (%)	3.07	3.31
Transfer	0.04	0.10

Table 6: Model Statistics under Labor Income Tax

	Top 10%	Top 20%	Top 30%	Top 40%	Top 50%	Gini Index
Wealth Share (%)						
Baseline	42	62	75	85	91	0.61
Labor tax	47	68	81	89	94	0.67
Income Share (%)						
Baseline	24	40	52	63	72	0.32
Labor tax	23	38	51	61	70	0.30

Table 7: Wealth and Income Distribution under Labor Tax

earnings of high-productivity workers and redistributing tax revenues to all households through lump-sum transfers, reducing the income Gini from 0.32 to 0.30. In summary, by increasing returns for existing asset holders through the general equilibrium channel, the introduction of a higher labor tax rate ultimately worsens wealth inequality while modestly improving income inequality.

3.2.2 Tax on Returns to Wealth

In this scenario, the government collects a 10% tax on earnings from asset holdings and then distributes the tax revenues equally to individuals in a lump-sum fashion. This tax on returns to wealth introduces a wealth tax wedge into the Euler equation, discouraging individuals from saving due to the lower marginal benefits of savings.

$$\text{participants: } u_1(c,l) = \beta \mathbb{E}_{s'|s}[u_1(c',l')\{1 + (1-\tau_w)(\underline{r} + r')\}] \tag{20}$$

$$\text{non-participants: } u_1(c,l) = \beta \mathbb{E}_{s'|s}[u_1(c',l')\{1 + (1-\tau_w)r_f\}] \tag{21}$$

The implementation of a tax on returns to wealth leads to a decrease in the aggregate capital stock by approximately 6%, from 3.44 to 3.22. Households tend to save less because the marginal benefits from saving decline. The overall capital stock reduction causes rental rates to rise and wages to fall. The increase in the common return component to risky assets (0.39 percentage points) through the general equilibrium effect partially offsets the subjective return loss caused by the 10% tax, but does not fully compensate for it. As a result, the wealth Gini rises by 1 percentage point from 0.61 to 0.62, while the income Gini falls by the same magnitude from 0.32 to 0.31. The income compression operates through two general equilibrium channels: the contraction in equilibrium wages (1.88 to 1.85) reduces the labor earnings of high-productivity workers relative to others, and the small increase in lump-sum transfers (0.04 to 0.05) financed by wealth tax revenue provides modest redistribution. In summary, a tax on returns to wealth fails to reduce wealth inequality despite a substantial 6% contraction in the capital stock—a notable inefficiency given the macroeconomic cost. The accompanying income compression of 1 percentage point reflects general equilibrium forces on wages and transfers.

Moment	Baseline	Wealth tax
Capital stock	3.44	3.22
Labor supply	0.36	0.35
Wage	1.88	1.85
Rental rate (%)	2.96	3.30
Common return for risky asset (%)	3.07	3.46
Transfer	0.04	0.05

Table 8: Model Statistics under Wealth Tax

	Top 10%	Top 20%	Top 30%	Top 40%	Top 50%	Gini Index
Wealth Share (%)						
Baseline	42	62	75	85	91	0.61
Wealth tax	42	63	76	85	92	0.62
Income Share (%)						
Baseline	24	40	52	63	72	0.32
Wealth tax	24	39	52	62	72	0.31

Table 9: Wealth and Income Distribution under Wealth Tax

Wealth taxation is thus ineffective at alleviating wealth inequality due to general equilibrium effects. However, if a progressive wealth tax were introduced instead of a constant tax rate, the outcomes would differ. Still, these results caution policymakers that taxes on asset returns could discourage capital accumulation in the economy.

3.2.3 Participation in Risky Asset Markets

We assume that government policy can effectively increase the market participation rate by 10 percentage points, from 60.7% to 70.7%. While this represents a substantial shift beyond Korea's historical range of homeownership rates, it is chosen to match the magnitude of the tax experiments and to provide a clear illustration of the participation channel. This increase may be achieved through policies that allow newly built houses to be sold exclusively to non-participant households, through new mortgage programs aimed at improving affordability for first-time homebuyers, or by introducing tax and regulatory measures that discourage concentrated ownership of multiple housing units.

Increased participation in risky asset markets has a minimal impact on aggregate economic moments: capital stock, labor supply, wages, and the rental rate remain essentially unchanged.

The wealth Gini index declines from 0.61 to 0.59 and the top 10% wealth share falls from 42% to 40%. The newly entered households accumulate assets more rapidly, thickening the middle class and thereby lowering the Gini coefficient. At the same time, the common return $\underline{r}$ declines from 3.07% to 2.97%, weakening the marginal saving incentives of the top segment. This decline arises from the mutual fund's market-clearing condition (Equation 15), under which the broader participant base shares the same pool of risky returns. Consequently, the combination of a diminished common return and a stronger middle class generates a meaningful reduction in wealth inequality. The income Gini index remains unchanged at 0.32, confirming that the participation channel operates primarily through wealth accumulation rather than labor earnings.

Moment	Baseline	Higher participation
Capital stock	3.44	3.44
Labor supply	0.36	0.36
Wage	1.88	1.88
Rental rate (%)	2.96	2.95
Common return for risky asset (%)	3.07	2.97
Transfer	0.04	0.04

Table 10: Model Statistics with Higher Market Participation

In summary, the implementation of a participation-expanding policy leads to a meaningful improvement in wealth inequality without any disturbance to macroeconomic conditions. This estimate should nonetheless be read in light of the model's abstraction from housing collateral and the leverage it permits, which would operate on several margins that need not push in the same direction. Households entering the market could lever

	Top 10%	Top 20%	Top 30%	Top 40%	Top 50%	Gini Index
Wealth Share (%)						
Baseline	42	62	75	85	91	0.61
Participation	40	60	74	83	90	0.59
Income Share (%)						
Baseline	24	40	52	63	72	0.32
Participation	24	39	52	63	72	0.32

Table 11: Wealth and Income Distribution with Higher Market Participation

up and accumulate risky wealth more rapidly, narrowing the gap to incumbent participants and amplifying the middle-class thickening that drives the Gini reduction documented above. The same mechanism, however, would widen the distance between new participants and the households that remain outside the market, so the net effect on aggregate concentration depends on the relative mass of these groups. Leverage would also scale each household's exposure to the idiosyncratic return process, amplifying dispersion within the participating group and thickening both tails of the wealth distribution, since levered positions magnify losses as well as gains. The price channel would differ as well. In our setting the risky asset is a pool of productive capital whose common return $\underline{r}$ falls as the participant base broadens, which weakens the saving incentives of the top segment; with housing in comparatively inelastic supply and mortgage credit expanding alongside participation, house prices could instead rise, delivering capital gains to incumbent owners and raising the cost of entry for new participants. The collateral channel therefore bears on the baseline wealth distribution as well as on the magnitude of the counterfactual, and a quantitative treatment of collateralized housing remains an important direction for future work.

3.2.4 Welfare Analysis

To evaluate the welfare implications of each policy, we compute the consumption-equivalent measure of welfare gain. This section focuses on the ex-ante welfare effects,[3] which measures the desirability of a policy change from the perspective of the initial steady state, before any transitions occur.

For each agent type (participants and non-participants), we first compute the consumption equivalent, $C_{eq} = u^{-1}((1-\beta)V)$, from their respective value functions, V. We

[3]The ex-ante consumption-equivalent variation answers the question: "What percentage of consumption would agents in the baseline economy be willing to give up (or need to receive) to be indifferent to a permanent switch to the new policy regime?"

then calculate the aggregated welfare gains between the baseline rules, $V0$, and the new policy rules, $V1$, using the baseline stationary distribution as weights. The aggregate and group-level welfare changes are the average percentage changes between these two values.

	Labor Tax (+10pp)	Wealth Tax (+10pp)	Participation (+10pp)
Total population	+0.14	+0.07	-0.02
Participants	+0.11	+0.06	-0.04
Non-participants	+0.20	+0.09	+0.01
Top 10% (by wealth)	+0.06	+0.02	-0.06
Bottom 50% (by wealth)	+0.32	+0.13	-0.00

Table 12: Consumption Equivalent Welfare Gains (%)

The results in Table 12 highlight a clear trade-off between equity and welfare. Both the labor income tax and the tax on returns to wealth generate positive welfare gains for the overall economy, with gains of 0.14% and 0.07%, respectively. These gains are driven by redistribution: the lump-sum transfers of tax revenue disproportionately benefit non-participants and the bottom 50% of the wealth distribution, whose welfare increases by 0.32% (labor tax) and 0.13% (wealth tax).

In contrast, the policy of expanding market participation generates a small aggregate welfare loss of -0.02%. The increased capital supply lowers the common return on risky assets, which adversely affects existing asset holders—participants lose 0.04% and the top 10% lose 0.06% in consumption-equivalent terms. Non-participants gain a negligible 0.01%, reflecting a slight wage increase from the larger capital stock. In an ex-ante sense, the losses to existing participants outweigh the gains to new entrants and non-participants, resulting in a small net welfare loss. From a purely utilitarian perspective, direct tax-and-transfer schemes are more effective at improving social welfare than expanding market access, even though the latter is more effective at reducing wealth inequality.

3.2.5 Policy Implications

Table 13 summarizes the effects of all three policy instruments under a uniform 10 percentage point shock. The policy experiments reveal three key findings. First, labor and wealth taxes are ineffective at alleviating wealth inequality. A higher labor income tax exacerbates wealth inequality through the general equilibrium channel, while a wealth tax on returns fails to reduce wealth inequality. This is primarily due to the general equilibrium effects of introducing a wedge in the optimal saving and labor conditions. Second, expanding access to risky asset markets effectively mitigates wealth inequality—reducing the Gini by 2 percentage points—without disrupting macroeconomic conditions. Third, there is a clear trade-off between equity and welfare: tax-and-transfer policies generate aggregate welfare gains through redistribution, whereas the participation policy improves equity at a small welfare cost to existing asset holders.

	Baseline	Labor tax	Wealth tax	Participation
Model Moments				
Capital stock	3.44	2.87	3.22	3.44
Labor supply	0.36	0.31	0.35	0.36
Wage	1.88	1.85	1.85	1.88
Common returns for risky asset (%)	3.07	3.31	3.46	2.97
Inequality				
Wealth Gini	0.61	0.67	0.62	0.59
Income Gini	0.32	0.30	0.31	0.32
Welfare				
Welfare (%)	—	+0.14	+0.07	-0.02

Table 13: Summary by Policy Instruments

As shown in Appendix D, the results are robust to the size of policy shocks: when the tax rates or participation rates are changed by only 1 percentage point, the qualitative implications remain virtually unchanged. This indicates that the findings are not sensitive to the magnitude of the policy experiment.

4 Conclusion

This paper has developed a parsimonious heterogeneous-agent general equilibrium model to study the drivers of wealth inequality in Korea. By combining limited participation in risky asset markets with persistent idiosyncratic returns to wealth, the model successfully replicates the highly skewed wealth and income distribution observed in Korean data. In particular, using the homeownership rate as a proxy for risky asset participation captures the central role of housing in shaping Korea's wealth inequality.

Policy experiments provide several key insights. First, labor and wealth taxes are ineffective policies for addressing wealth inequality. A higher labor income tax exacerbates wealth inequality through the general equilibrium channel—by raising returns to capital and penalizing labor income earners who have not yet accumulated wealth—though it modestly reduces income inequality. A tax on returns to wealth fails to alleviate wealth inequality, as the general equilibrium increase in the common risky return partially offsets the direct redistributive effect. Second, promoting participation in risky asset markets effectively alleviates wealth inequality—reducing the wealth Gini by 2 percentage points—without distorting macroeconomic stability. Third, there is a fundamental trade-off between equity and welfare: tax-and-transfer schemes generate aggregate welfare gains through redistribution, while the participation policy improves equity at a small welfare cost to existing asset holders.

While these findings are encouraging, several limitations remain. Most importantly, the model proxies risky asset participation by homeownership and treats pooled housing savings as productive capital. This treatment has precedent in the macroeconomic literature, including the basic model of Iacoviello (2005) and the productive-land framework of Liu et al. (2013), but it abstracts from housing-specific features—illiquidity, mortgage leverage, and collateral constraints—that shape wealth accumulation among homeowners. As discussed in Section 3, the collateral channel would operate on several offsetting margins and bears on the baseline wealth distribution as well as on our counterfactual estimates. In addition, participation is modeled exogenously. In reality, households endogenously choose whether to enter the housing market in response to financial frictions such as down-payment requirements and credit limits; the stability of Korea's homeownership rate over the past eight years makes the fixed-participation approximation reasonable for long-run analysis, but it precludes an assessment of how policies alter participation incentives at the margin.

Extending the model to include endogenous participation decisions and explicit housing frictions would provide more precise insights into how policy measures alter households' incentives to invest and accumulate wealth. Similarly, while this paper assumes an exogenous idiosyncratic return process, future work could incorporate endogenous return heterogeneity arising from portfolio choice, entrepreneurial investment, or heterogeneous access to financial intermediaries. Such extensions would generate richer implications for both inequality and aggregate dynamics. We leave these important extensions for future research.

Overall, this paper highlights the importance of financial market participation margins and return heterogeneity in understanding the Korean wealth distribution. The results underscore that policies expanding participation in risky asset markets can effectively improve equity, while tax-and-transfer schemes remain more effective for enhancing aggregate welfare through redistribution.

References

Aiyagari, S. R. (1994). Uninsured idiosyncratic risk and aggregate saving. *The Quarterly Journal of Economics 109*(3), 659–684.

Bach, L., L. E. Calvet, and P. Sodini (2020). Rich pickings? risk, return, and skill in household wealth. *American Economic Review 110*(9), 2703–2747.

Basak, S. and D. Cuoco (1998). An equilibrium model with restricted stock market participation. *The Review of Financial Studies 11*(2), 309–341.

Benhabib, J., A. Bisin, and M. Luo (2019). Wealth distribution and social mobility in the US: A quantitative approach. *American Economic Review 109*(5), 1623–1647.

Bewley, T. (1977). The permanent income hypothesis: A theoretical formulation. *Journal of Economic Theory 16*, 252–292.

Causa, O., N. Woloszko, and D. Leite (2019). Housing, wealth accumulation and wealth distribution: Evidence and stylized facts. *OECD Economics Department Working Papers* (1588).

Fagereng, A., L. Guiso, D. Malacrino, and L. Pistaferri (2020). Heterogeneity and persistence in returns to wealth. *Econometrica 88*, 115–170.

Ferraro, D. and V. Valaitis (2025). Consumption quality and employment across the wealth distribution. *The Review of Economic Studies 92*(3), 1801–1836.

Greenwood, J., Z. Hercowitz, and G. W. Huffman (1988). Investment, capacity utilization, and the real business cycle. *The American Economic Review 78*(3), 402–417.

Guvenen, F. (2009). A parsimonious macroeconomic model for asset pricing. *Econometrica 77*(6), 1711–1750.

Hubmer, J., P. Krusell, and A. A. Smith (2021). Sources of US wealth inequality: Past, present, and future. *NBER Macroeconomics Annual 35*, 391–455.

Huggett, M. (1993). The risk-free rate in heterogeneous-agent incomplete-insurance economies. *Journal of Economic Dynamics and Control 17*, 953–969.

Iacoviello, M. (2005). House prices, borrowing constraints, and monetary policy in the business cycle. *American Economic Review 95*(3), 739–764.

Krusell, P. and A. A. Smith (1998). Income and wealth heterogeneity in the macroeconomy. *Journal of Political Economy 106*(5), 867–896.

Liu, Z., P. Wang, and T. Zha (2013). Land-price dynamics and macroeconomic fluctuations. *Econometrica 81*(3), 1147–1184.

Ministry of Economy and Finance (2025). The third long-term fiscal projection (2025–2065). Technical report, Ministry of Economy and Finance, Republic of Korea.

Nam, S. (2015). The characteristics of Korea's household income and wealth distributions. Health and Welfare Issue & Focus 277, Korea Institute for Health and Social Affairs.

Park, J., S. Lee, and E. Tak (2024). Fluctuations and economic inequality - focusing on asset and income inequality. *The Korean Journal of Economics 31*(1), 59–90.

Tauchen, G. (1986). Finite state markov-chain approximations to univariate and vector autoregressions. *Economics Letters 20*(2), 177–181.

Xavier, I. (2021). Wealth inequality in the US: the role of heterogeneous returns. Working paper.

Young, E. R. (2010). Solving the incomplete markets model with aggregate uncertainty using the Krusell–Smith algorithm and non-stochastic simulations. *Journal of Economic Dynamics and Control 34*, 36–41.

Wealth Inequality in Korea:
Limited Participation and Idiosyncratic Returns

—Supplemental Material—

A Numerical Solution

We solve the model following the standard approach of heterogeneous-agent Aiyagari-type models. The household problems are solved by value function iteration under discretized state spaces for assets, labor productivity, and idiosyncratic returns. The idiosyncratic shocks to labor productivity and returns are approximated by finite-state Markov chains constructed from the underlying AR(1) processes by following Tauchen (1986). Since the household's optimal savings choice is generally not located exactly on the asset grid, we employ off-grid search using the Golden Section Search method with cubic spline interpolation.

We use a log-spaced grid with 1000 points for risky-asset holdings with an upper bound of 1000 (about one hundred fifty times aggregate capital stock) to minimize truncation errors. A log-spaced grid with 200 points for the risk-free asset with an upper bound of 10 is used in the model. In addition, five thousand equally spaced points are used for the distribution of asset holdings. Both labor productivity and return shocks take 13 states, respectively. Changes in the number and bounds of grids do not have a significant impact on the results of models.

Convergence of the value function iteration is determined by the sup norm with a tolerance of 10^{-6}. The distribution of households is updated using the endogenous policy rules and the Markov transition matrices until a stationary distribution is obtained. General equilibrium is achieved by iterating on factor prices until both the capital and labor markets clear simultaneously. The model is implemented using the Intel Fortran programming language and incorporates OpenMP for parallelization.

The detailed computation algorithm of the model follows these steps:

1. Start with an initial guess for the aggregate capital stock and labor.

2. Given the guessed values for aggregate capital stock K and labor L, determine the rental rates r^* and wage w using Equations (12) and (13). Additionally, make an initial guess for the common return component $\underline{r}$.

(a) Solve the problems of the two types of individuals, Equations (6) and (9), given the rental rates, wage, and common return component. This can be done using the golden section search method with cubic spline interpolation.

(b) Simulate the distribution of participants μ_p and non-participants μ_n using their respective decision rules and the transition matrices of labor productivity and return shocks, following the simulation method outlined by Young (2010).

(c) Using the distributions of participants and non-participants, check if the aggregate returns from risky assets and risk-free assets are sufficiently close to the rental revenue of mutual funds. If not, return to step (a) and make a new guess for the common return component.

(d) With the distributions of participants and non-participants, calculate the aggregate capital and labor. If the calculated aggregate capital and labor are sufficiently close to the initial guesses, the solution algorithm is complete. If they are not close enough, return to step 1 to make a new guess for aggregate capital and labor.

B Data

We collect data on wealth distribution and income distribution in Korea from the 2024 Survey of Household Finances and Living Conditions (SFLC), conducted by Statistics Korea, the Financial Supervisory Service, and the Bank of Korea. Specifically, we use the 2024 wealth distribution and the 2023 income distribution from 2024 SFLC. For income distribution, we use data based on disposable income, including labor income, business income, asset income, private net transfers, and public net transfers.

Because the published SFLC statistics do not report top 1% and top 5% wealth shares, these top-tail moments are not targeted in the calibration. To assess how well the model captures the upper tail, we compare the model-implied shares against an independent estimate from Nam (2015), who computes top-tail statistics directly from the 2014 Household Finance and Welfare Survey microdata. The 2014 figures are 12%, 30%, and 44% for the top 1%, 5%, and 10% of households in wealth, against which the model generates 9%, 27%, and 42%. The gap between model and data is roughly 2–3 percentage points and is essentially the same at the untargeted top 1% and top 5% as at the targeted top 10%, indicating that the model captures the shape of the upper tail rather than merely matching the targeted decile. This is a non-trivial outcome since the wealth-inequality literature has long emphasized the difficulty of matching top shares in workhorse heterogeneous-agent

	Wealth Inequality		Income Inequality	
	Data (2024)	Model	Data (2023)	Model
Top 1%	—	9	—	4
Top 5%	—	27	—	14
Top 10%	44	42	24	24
Top 20%	63	62	39	40
Top 30%	75	75	52	52
Top 40%	84	85	63	63
Top 50%	90	91	72	72
Top 60%	95	95	80	80
Top 70%	98	98	87	87
Top 80%	99	99	93	92
Top 90%	100	100	98	97
Top 100%	100	100	100	100
Gini index	0.61	0.61	0.32	0.32

Table A.1: Wealth and Income Distribution in Korea

models. The corresponding income comparison is also reproduced well: the model–data gaps in the top income shares (5, 4, and 3 percentage points at the top 10%, 5%, and 1%) are comparable to those for wealth and do not widen at the top.

Data on homeownership rates in Korea are collected from the 2024 Korea Housing Survey. Homeownership rates in Korea have remained stable over the past eight years.

	2016	2017	2018	2019	2020	2021	2022	2023
Homeownership rates	59.9	61.1	61.1	61.2	60.6	60.6	61.3	60.7

Table A.2: Homeownership Rates in Korea (%)

C Decomposition of Wealth and Income Concentration by Channel

The model combines three sources of concentration: uninsurable idiosyncratic income risk, persistent idiosyncratic returns to wealth, and limited participation in risky asset markets. To assess the quantitative contribution of each mechanism, and in particular to verify that the return process is not silently absorbing variation that the other channels should explain, Table A.3 reports the model's wealth distribution as the return-heterogeneity and limited-participation channels are removed in turn. Removing return heterogeneity sets $\sigma_r = 0$, so that all risky-asset holders earn the common return $\underline{r}$; removing limited participation sets $\lambda = 1$, so that all households access the risky asset. Eliminating both leaves the standard uninsurable-income-risk economy in the tradition of Aiyagari (1994) and Krusell and Smith (1998).

	Top 10%	Top 30%	Top 50%	Wealth Gini
Data	44	75	90	0.61
Baseline (all channels)	42	75	91	0.61
without return heterogeneity	36	72	90	0.57
without limited participation	36	70	88	0.55
income risk only (both removed)	32	66	85	0.51

Table A.3: Decomposition of Wealth Concentration by Channel

Three features stand out. First, the uninsurable-income-risk-only economy generates a top 10% wealth share of 32% and a wealth Gini of 0.51. This level of concentration reflects the substantial income dispersion required to match Korea's income Gini of 0.32; even so, it falls well short of the 44% top 10% share observed in the data, consistent with the long-standing finding that workhorse heterogeneous-agent models understate top wealth concentration.

Second, adding return heterogeneity and limited participation raises the top 10% share from 32% to 42%, closing most of the gap between the workhorse benchmark and the data. The two channels contribute almost symmetrically: removing either one lowers the top 10% share by 6 percentage points (from 42% to 36% in both cases) and the wealth Gini by 0.04 and 0.06 points, respectively.

Third, the two channels contribute largely independently. Measured from the full model, removing return heterogeneity lowers the wealth Gini by 0.04 and removing limited participation by 0.06; removing both lowers it by 0.10, exactly the sum, so the Gini exhibits no interaction. The top-share moments display only small interactions: the effect is slightly sub-additive at the very top—the top 10% share falls by 6 percentage points when either channel is removed but by 10 rather than 12 when both are—and slightly super-additive lower in the distribution, where the combined effect exceeds the sum of the individual effects by 1 and 2 percentage points at the top 30% and top 50% shares, respectively. In every case the interaction is small relative to the direct contributions, so the substantive conclusion is robust: each channel makes a large and largely independent contribution to wealth concentration, and neither the return process nor the participation split is silently absorbing the role of the other.

	Top 10%	Top 30%	Top 50%	Income Gini
Data	24	52	72	0.323
Baseline (all channels)	24	52	72	0.318
without return heterogeneity	24	52	72	0.312
without limited participation	23	52	71	0.309
income risk only (both removed)	23	51	71	0.306

Table A.4: Decomposition of Income Concentration by Channel

Table A.4 repeats the exercise for the income distribution, and the contrast with wealth is stark. Income concentration is almost invariant to the two channels: the income Gini moves only from 0.318 to 0.306 as both are removed, and the top 10% income share barely changes, from 24% to 23%. This pattern reflects the structure of the calibration. The persistence and volatility of the labor-productivity process, ρ_e and σ_e, are disciplined by the income moments and govern the income distribution; the return parameters and the participation margin are disciplined by, and specialize in, the wealth distribution. Removing the latter two therefore leaves income concentration essentially untouched while substantially lowering wealth concentration. This separation confirms that the two wealth-specific channels operate through wealth accumulation rather than labor earnings—consistent with the participation experiment in the main text, where the income Gini was unchanged—and that the parameters governing each margin are not contaminating the moments they are not meant to explain.

D Robustness Checks for Policy Implications

Table A.5 and Table A.6 present robustness checks based on a one-percentage-point increase in taxes and a one-percentage-point increase in participation rates in risky markets. While the quantitative effects are small in magnitude, the qualitative patterns remain consistent with the main analysis: both labor and wealth taxes are ineffective policies for mitigating wealth inequality, while a participation-expanding policy reduces wealth inequality without affecting macroeconomic conditions. These results confirm that the main conclusions are not influenced by the size of the shock.

Moment	Baseline	Labor tax	Wealth tax	Participation
Capital stock	3.44	3.37	3.41	3.44
Labor supply	0.36	0.35	0.35	0.36
Wage	1.88	1.88	1.88	1.88
Rental rate (%)	2.96	3.00	2.99	2.96
Common return for risky asset (%)	3.07	3.10	3.10	3.06

Table A.5: Model Statistics under Marginal Policy Changes (+1pp)

	Top 10%	Top 20%	Top 30%	Top 40%	Top 50%	Gini Index
Wealth Share (%)						
Data	44	63	75	84	90	0.612
Baseline	42	62	76	85	91	0.612
Labor tax (+1pp)	43	63	76	85	92	0.619
Wealth tax (+1pp)	42	62	76	85	91	0.612
Participation (+1pp)	41	62	76	85	91	0.610
Income Share (%)						
Data	24	40	52	63	72	0.323
Baseline	24	40	52	63	72	0.318
Labor tax (+1pp)	24	39	52	62	72	0.315
Wealth tax (+1pp)	24	40	52	63	72	0.317
Participation (+1pp)	24	40	52	63	72	0.317

Table A.6: Wealth and Income Inequality under Marginal Policy Changes

E Sensitivity to the Persistence of Idiosyncratic Returns

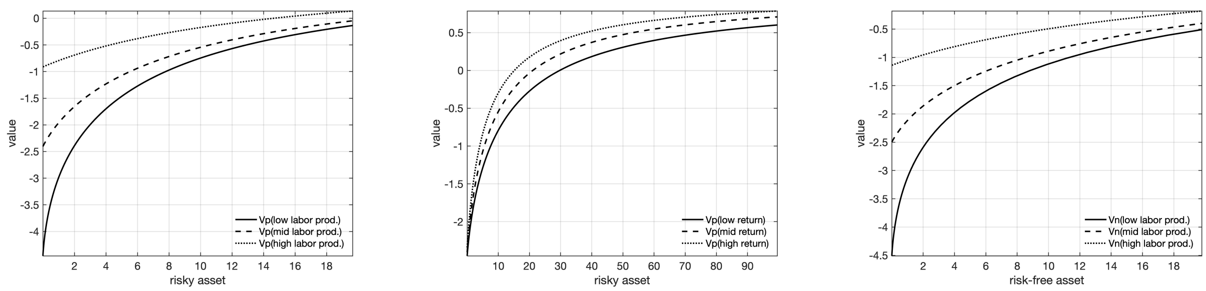
To verify that the calibrated persistence of idiosyncratic returns is disciplined by the data rather than chosen freely, Table A.7 reports the model's wealth distribution under alternative values of ρ_r, holding all other parameters at their baseline values. As ρ_r falls below its calibrated value of 0.97, the model's top wealth shares decline and move away from the data targets, confirming that high return persistence is required to replicate the observed concentration of wealth. This addresses the concern that the return parameters may absorb variation that a richer model would attribute to other channels: the persistence is pinned down by the wealth distribution moments, not assigned arbitrarily.

	Top 10%	Top 20%	Top 30%	Top 40%	Top 50%	Gini Index
Data	44	63	75	84	90	0.61
Baseline ($\rho_r = 0.97$)	42	62	75	85	91	0.61
$\rho_r = 0.95$	39	60	74	84	90	0.59
$\rho_r = 0.90$	37	58	73	83	90	0.57

Table A.7: Wealth Distribution under Alternative Return Persistence

F Model Results

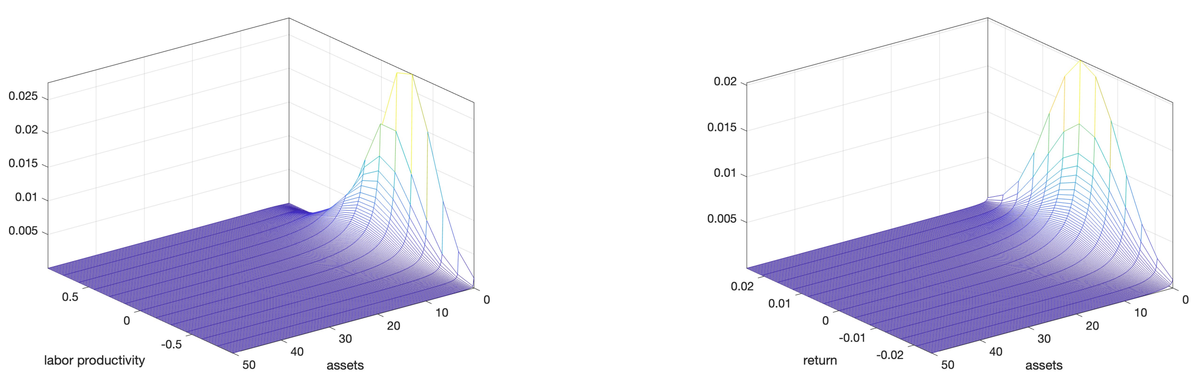
In this section, we present detailed results of the model and distribution. Figure A.1 describes the values of participants and non-participants. Figure A.2 describes the distribution of population. Figure A.3 describes the wealth and income distribution of the model economy.



(a) Value of Participants (b) Value of Participants (c) Value of Non-participants

Figure A.1: Value Functions

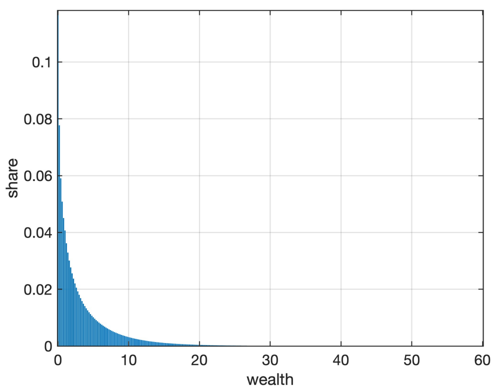
NOTES: Subfigure (a) illustrates the value of participants across labor productivities with a medium idiosyncratic return shock. Subfigure (b) illustrates the value of participants across idiosyncratic returns with a medium labor productivity shock. We truncate the asset-holding grids for a better appearance because value functions are flat in areas of large asset holdings.



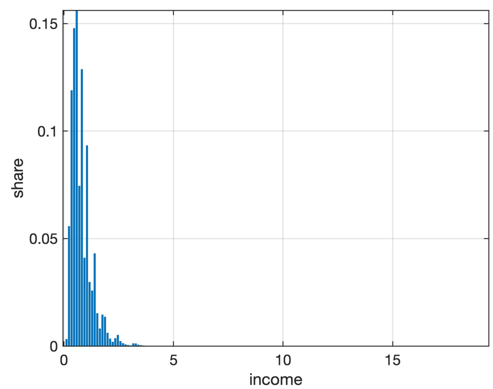
(a) Labor Productivity and Asset Holdings (b) Return and Asset Holdings

Figure A.2: Distribution

NOTES: To improve the appearance of figures, we truncate the asset-holding grids since the population density is very low in the upper tail of the wealth distribution.



(a) Wealth Distribution



(b) Income Distribution

Figure A.3: Wealth and Income Distribution

NOTES: To improve the appearance of figures, we truncate asset holding grids since the population density is very low in the upper tail of the wealth distribution.